Familiarizzazione con TIAGo

Fondamenti di Robotica

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Overview del robot TIAGo

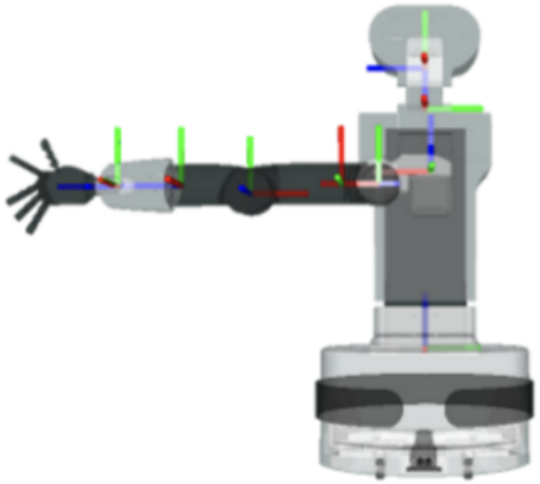
Componenti principali



Specifiche

Dimensions	Height	110 – 145 cm
	Weight	72 Kg
	Base footprint	Ø 54 cm
Degrees of freedom	Mobile base	2
	Torso lift	1
	Arm	4
	Wrist	3
	Head	2
	Hey5 hand	19 (3 actuated)
	PAL gripper	2
Mobile base	Drive system	Differential
	Max speed	1 m/s
Torso	Lift stroke	35 cm
Arm	Payload	2 Kg
	Reach	87 cm
Electrical features	Battery	36 V, 20 Ah
Sensors	Base	Laser range-finder
		Sonars
		IMU
	Torso	Stereo microphones
	Arm	Motors current feedback
	Wrist	Force/Torque
	Head	RGB-D camera

Specifiche cinematiche del TIAGo

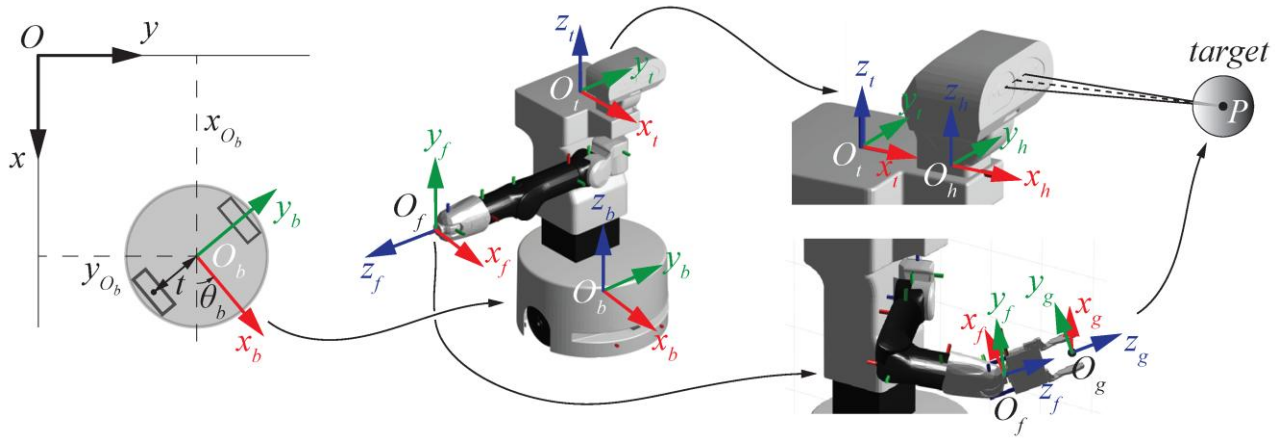


Joint	Type	Lower limit	Upper limit
torso	prismatic	0 mm	350 mm
head_1	revolute	-75º	75º
head_2	revolute	-60º	45º
arm_1	revolute	0	157.5º
arm_2	revolute	-90º	62.5º
arm_3	revolute	-202.5º	90º
arm_4	revolute	-22.5º	135º
arm_5	revolute	-120º	120º
arm_6	revolute	-90º (*)	90º (*)
arm_7	revolute	-120º	120º

(*) Wrist version with force/torque sensor arm_6 range is [-81º, 81º]

Sistemi di riferimento e cinematica del TIAGo

Sistemi di riferimento



$O_b - x_b y_b z_b \rightarrow$ riferimento della piattaforma mobile

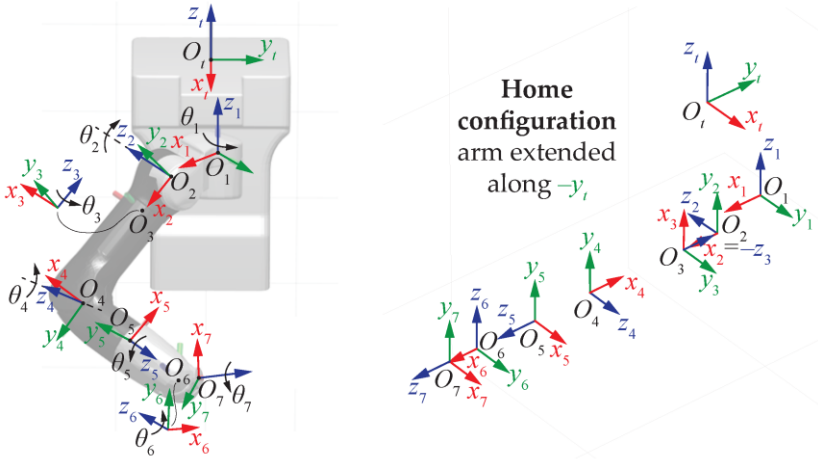
$O_t - x_t y_t z_t \rightarrow$ riferimento del tronco

$O_h - x_h y_h z_h \rightarrow$ riferimento della base della testa

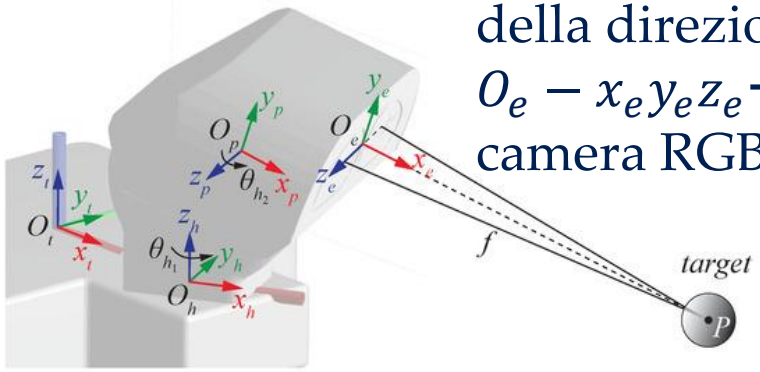
$O_f - x_f y_f z_f \rightarrow$ riferimento dell'ee del braccio robotico

$O_g - x_g y_g z_g \rightarrow$ riferimento del gripper

Cinematica braccio



Cinematica testa



$O_p - x_p y_p z_p \rightarrow$ riferimento responsabile della direzione di puntamento della testa

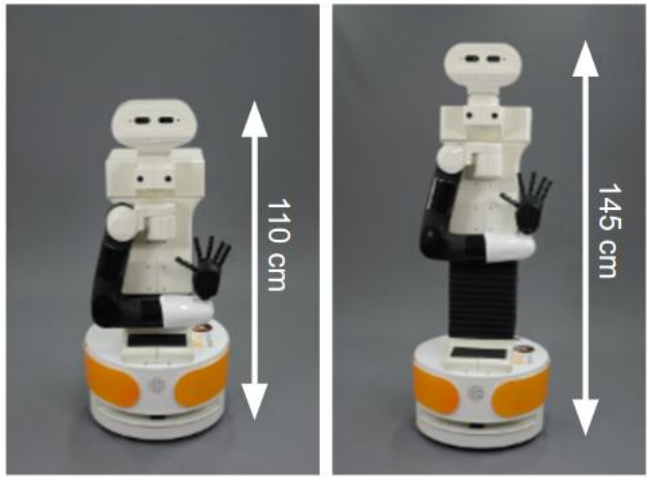
$O_e - x_e y_e z_e \rightarrow$ riferimento dei sensori della camera RGB

NB: i movimenti avvengono attorno all'asse z dei sistemi di riferimento considerati

Bajrami, A.; Palpacelli, M.-C.; Carbonari, L.; Costa, D. Posture Optimization of the TIAGo Highly-Redundant Robot for Grasping Operation. Robotics 2024, 13, 56. https://doi.org/10.3390/robotics13040056

Hardware del TIAGo

Torso

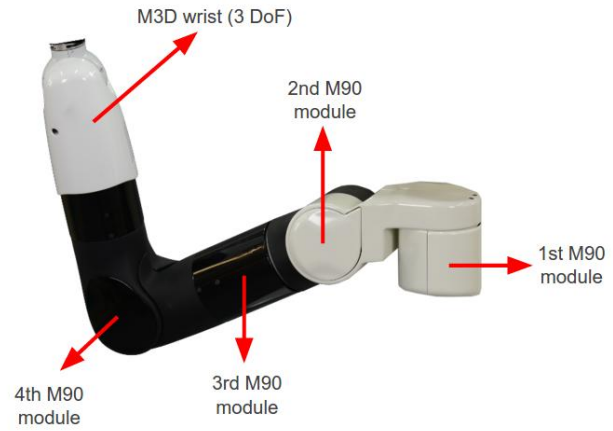


Struttura che supporta braccio e testa

- v = 50 mm/s
- Corsa = 350 mm

Weight	10 Kg					
Payload	2.8 Kg					
Joints	7					
Onboard control modes	Modules	Position, velocity and current				
	Wrist	Position and velocity				
Actuators					Encoders (bits)	
	Description	Reduction	Max speed [rpm]	Nominal torque [Nm]	Motor	Absolute
	1st module	100:1	18	39	12	12
	2nd module	100:1	18	39	12	12
	3rd module	100:1	22	22	12	12
	4th module	100:1	22	22	12	12
	Wrist 1st DoF	336:1	17	3	11	12
	Wrist 2nd DoF	336:1	17	5	11	13
	Wrist 3rd DoF	336:1	17	5	11	13

Braccio



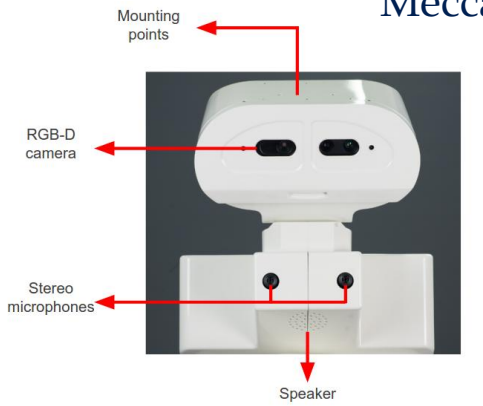
Parallel Gripper

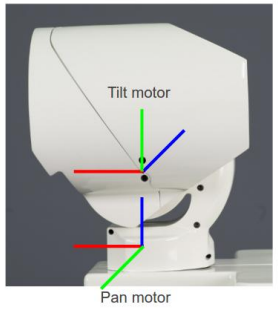
- Range 8 cm (4 cm per lato)



Testa

Meccanismo pan-tilt (2 DoF)





RGB-D main specifications ¶	
Manufacturer	Orbbec
Model	Astra S
Field of view	60◦H, 49.5◦V, 73◦D
Interface	USB 2.0
Color stream modes	QVGA 320×240 @ 30 fps, VGA 640×480 @ 30 fps, 1280×960 @ 10 fps
Depth stream modes	QVGA 320×240 @ 30 fps, VGA 640×480 @ 30 fps, 160×120 @ 30 fps
Depth sensor range	0.4 - 2 m

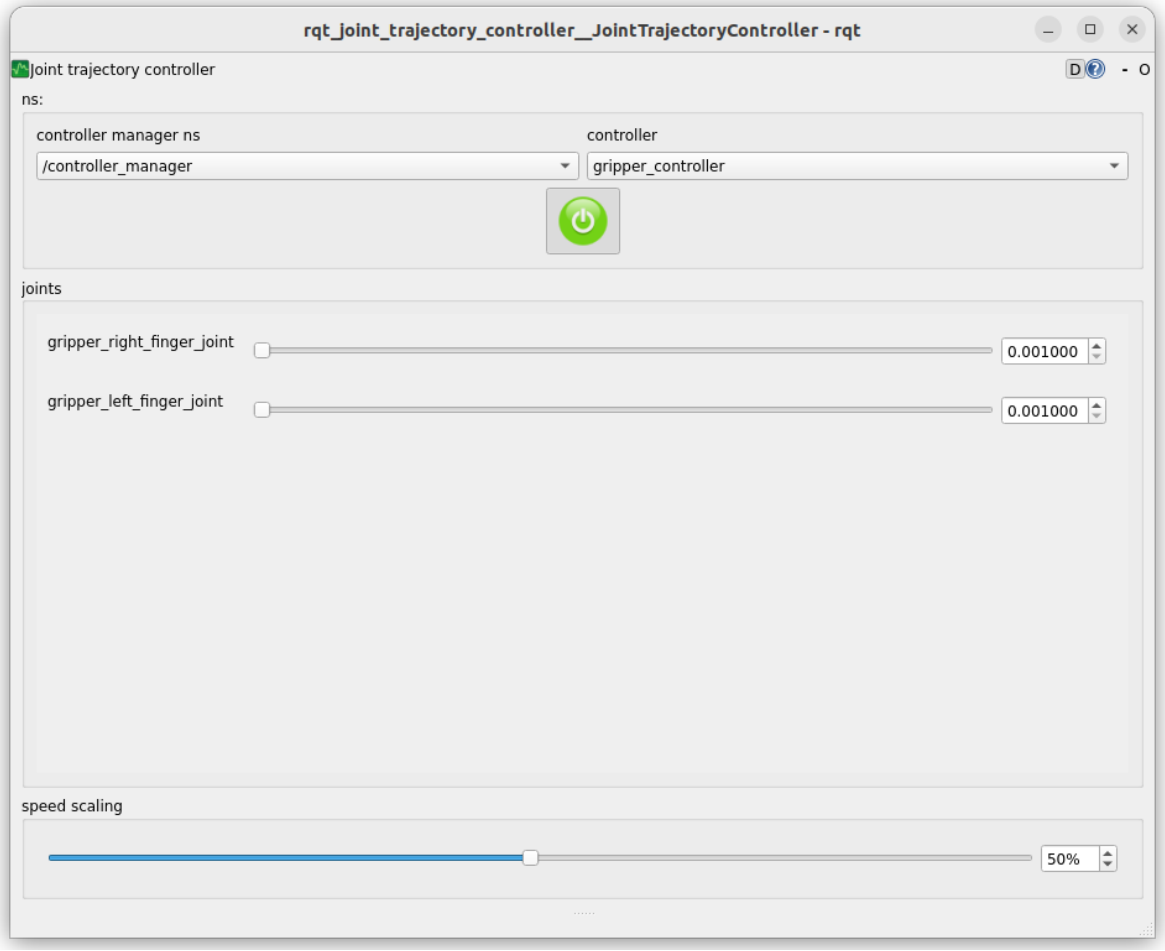
Installazioni

```
sudo apt update
sudo apt install ros-humble-rqt-joint-trajectory-controller
sudo apt install python3-pip
```

Controllo end-effector

I giunti dell'end-effector possono essere mossi individualmente usando una GUI ROS

```
ros2 run rqt_joint_trajectory_controller rqt_joint_trajectory_controller
```



Tramite GUI si possono muovere:

- Testa
- Torso
- Gripper
- Arm

- ❑ Il gruppo di giunti da muovere va selezionato dalla lista «controller»
- ❑ Cliccare su pulsante di avvio e muovere i cursori
- ❑ Vedere movimento su Gazebo

Topic utili per progetto

ros2 topic list

- /arm_controller/joint_trajectory --> Moves the arm joints to the desired configuration
- /arm_controller/state --> To know the actual values of arm joints
- /gripper_controller/joint_trajectory --> Moves the gripper joints to the desired configuration
- /gripper_controller/state --> To know the actual values of gripper joints
- /head_controller/joint_trajectory --> Moves the head joints to the desired configuration
- /head_controller/state --> To know the actual values of head joints
- /head_front_camera/rgb/camera_info --> Gives the camera calibration
- /head_front_camera/rgb/image_raw --> Streams the camera raw images
- /joint_states --> To know all joints
- /robot_description --> Provides the robot URDF model
- /tf --> Broadcasts coordinate frames
- /torso_controller/joint_trajectory --> Moves the torso joint to the desired configuration
- /torso_controller/state --> To know the actual values of torso joint

Informazioni utili

ros2 interface show trajectory_msgs/msg/JointTrajectory

ros2 topic info /head_controller/joint_trajectory

ros2 topic echo /head_controller/state

ros2 control list_controllers

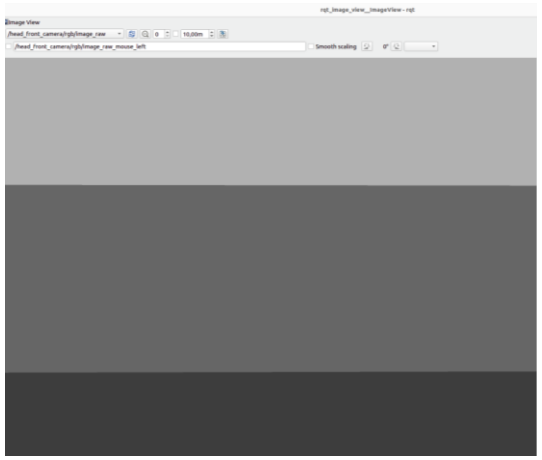
ros2 topic echo /joint_states

ros2 param get /head_controller joints

Visuale camera

Ciò che vede il TIAGo può essere visto tramite

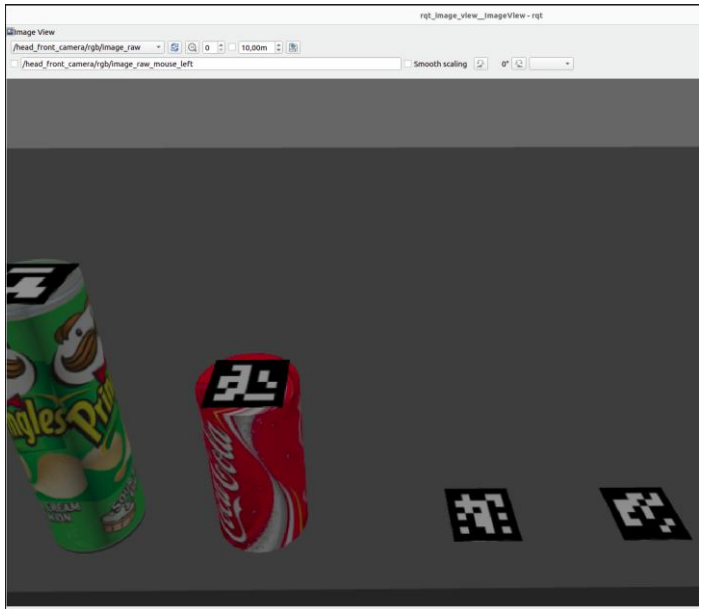
```
ros2 run rqt_image_view rqt_image_view
```



Tramite GUI selezionare:

- /head_front_camera/rgb/image_raw

```
ros2 topic pub /head_controller/joint_trajectory trajectory_msgs/msg/JointTrajectory "{
  joint_names: ['head_1_joint', 'head_2_joint'],
  points: [
    {
      positions: [6.67e-07, -0.86],
      time_from_start: {sec: 2}
    }
  ]
}"
```



Action utili per progetto

ros2 action list

- /arm_controller/follow_joint_trajectory → Moves the arm joints to the desired positions
- /gripper_controller/follow_joint_trajectory →This action encapsulates the trajectory_msgs/JointTrajectory message
- /head_controller/follow_joint_trajectory → Moves the head joints to the desired positions
- /head_controller/point_head_action → Moves the head joints to the necessary positions to point to a desired position, performing inverse kinematics
- /torso_controller/follow_joint_trajectory → This action encapsulates the trajectory_msgs/JointTrajectory message

Comandi utili per familiarizzare con il robot (Task 1)

Visualizzare controlli attivi

ros2 control list_controllers

Visualizzare stato dei giunti

ros2 topic echo /joint_states

Visualizzare stringa dei giunti in head_controller

ros2 param get /head_controller joints

Aprire la GUI per Rviz

ros2 run joint_state_publisher_gui joint_state_publisher_gui

Aprire la telecamera del TIAGo

ros2 run rqt_image_view rqt_image_view → settare il topic corretto dalla GUI della camera (head_front_camera/rgb/image_raw)

Spostare la testa del TIAGo da terminale

ros2 topic pub /head_controller/joint_trajectory trajectory_msgs/msg/JointTrajectory "{ joint_names: [‘head_1_joint’, ‘head_2_joint'], points: [{positions: [-2.17e-05, -0.57], time_from_start: {sec: 2}}]} "

Esercitazione

- Rilevamento oggetti tramite ArUco

→ Publisher, subscriber, tf, action, service

